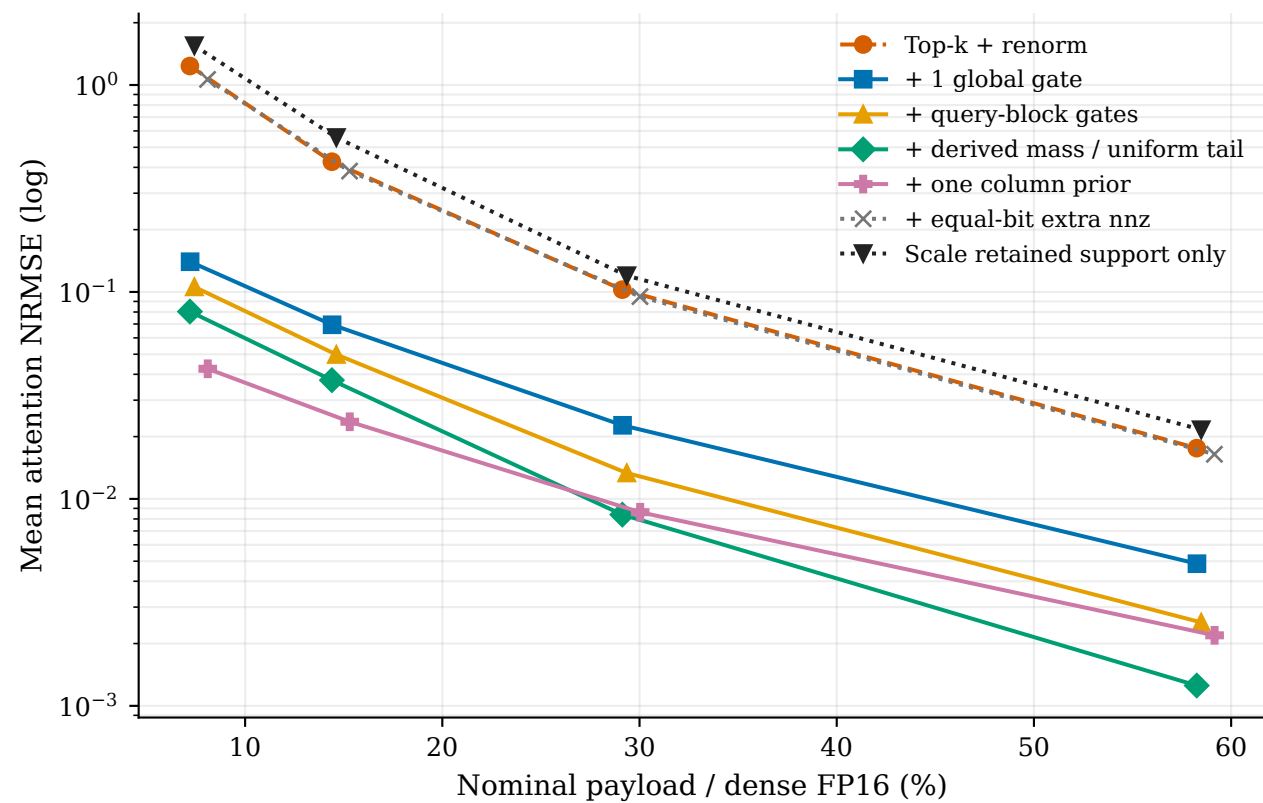
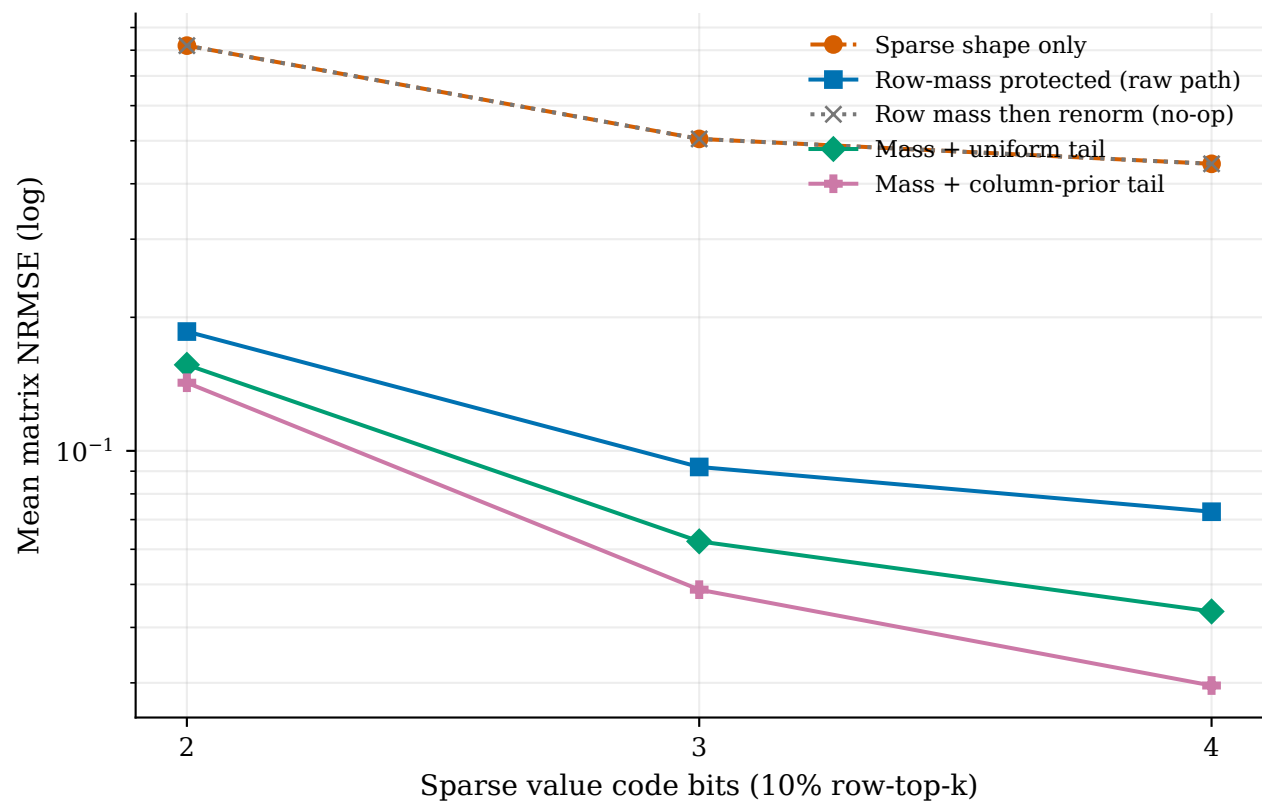
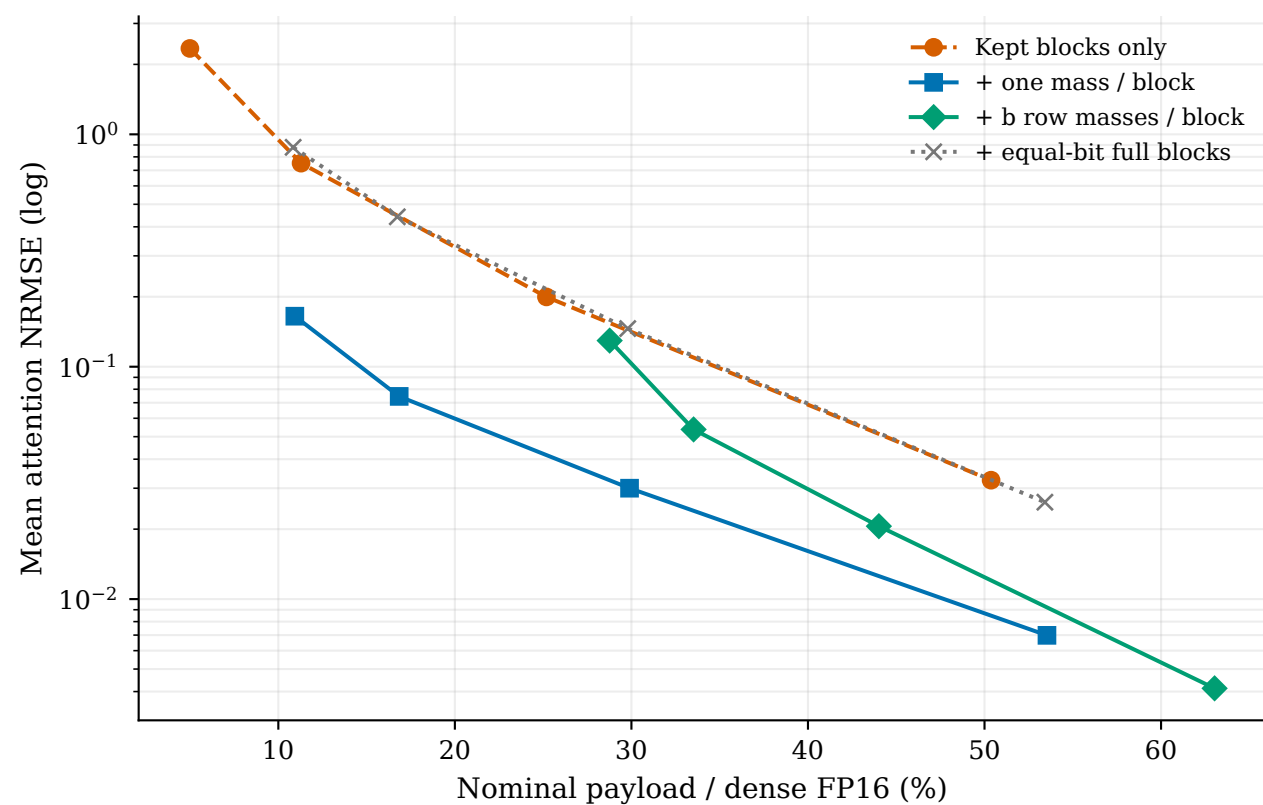
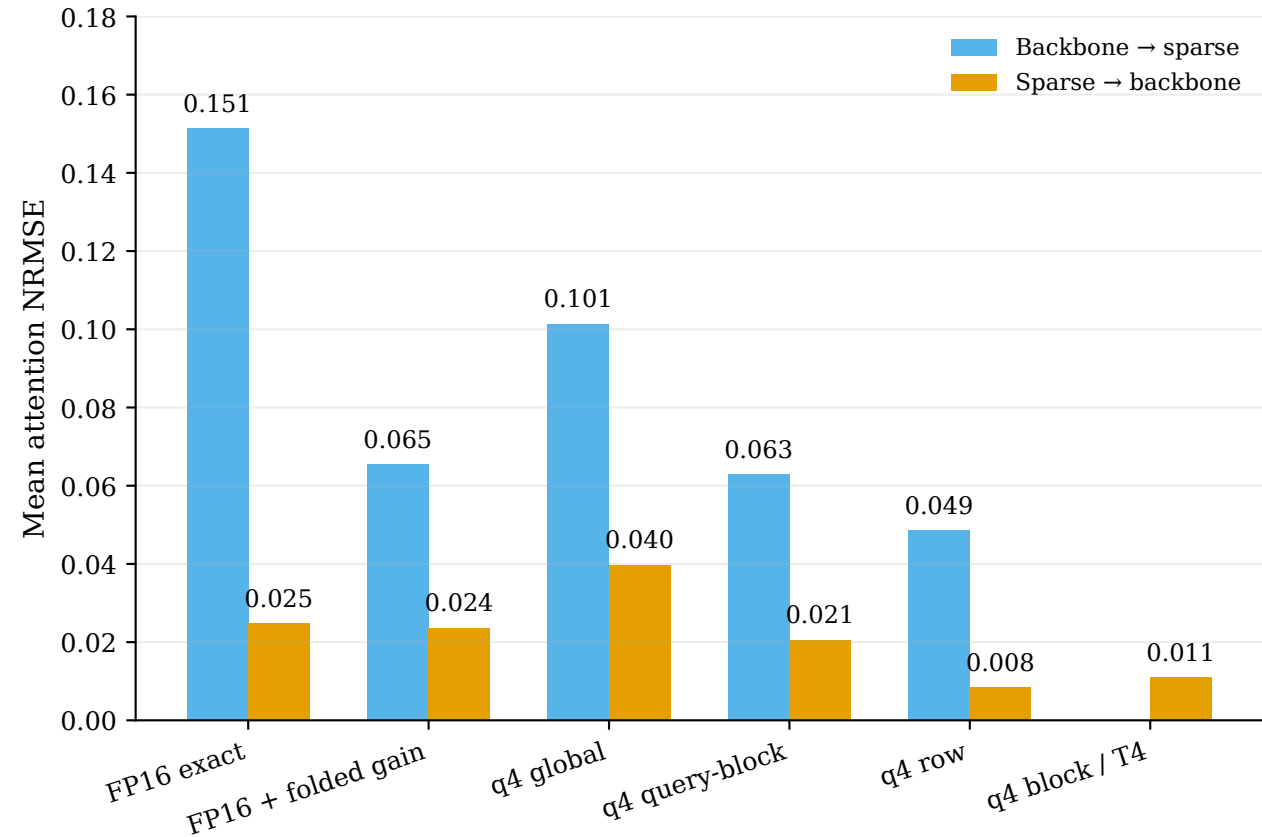
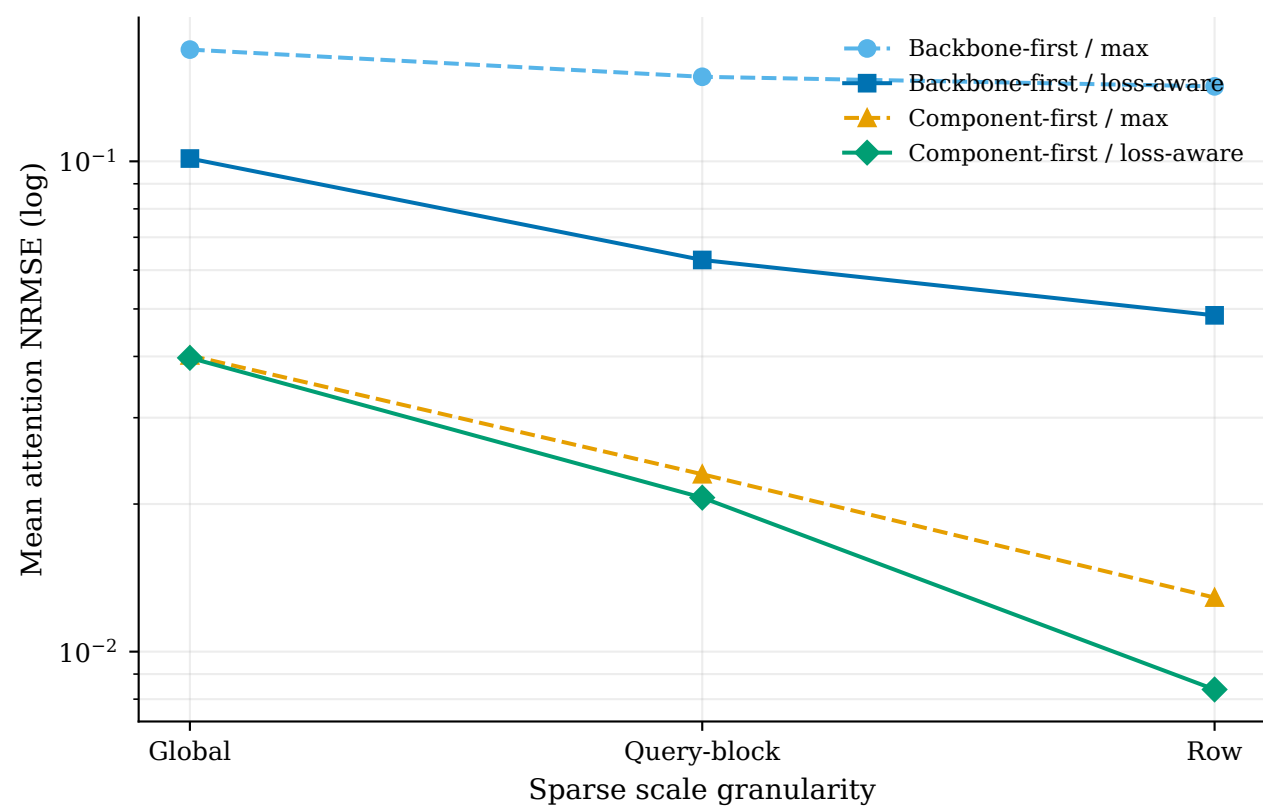


(a) Row-top-k: protect missing probability mass**(b) Mass scale helps raw amplitude; tail repair survives renorm****(c) Block pruning: mass summaries beat equal-bit extra blocks****(d) Same 25% cap: outlier-first and fine scales win****(e) Same bits: optimize scale for the reported loss****(f) Parameter envelope remains target-fitted**